Workshop on Embodied Agentic Systems
for Urban and Environmental Observation





Multi-Agentic Systems for Observational Sciences

Agents Reducing Barriers to Using Urban and Environmental Instruments

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PART 1 · THE SYSTEMS

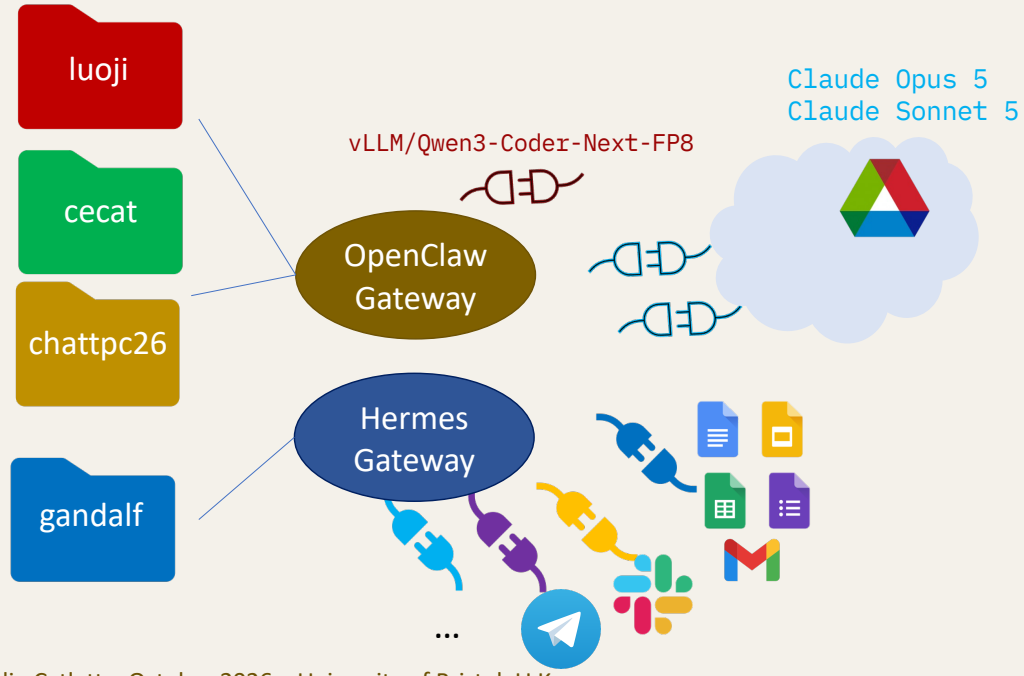
Agentic Frameworks: OpenClaw, Hermes

≠ chatbot

Waits for a prompt, answers, and idles until the next one.

≠ agent

Acts *between* conversations — on a schedule, on events, on its own judgment.



OpenClaw, Hermes

Frameworks that *run* agents: a **gateway**, **plugins** and channels, and per-agent **markdown files** that define each agent's identity, role, and behavior.

Model routing & fallback

Agents/Gateways run on a NVIDIA DGX-Spark and reach remote models at Argonne — falling back to local `LLM` if remote model access fails or is interrupted.

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PART 1 · THE SYSTEM

My Deployment: Three Agents

Multiple agents, multiple roles — running a major conference, managing my personal gmail, and overall supervision.

luoji

OpenClaw

Supervisor

Reviews outbox and slack drafts, handles administrative queries. The reviewer in the loop.

cecat

OpenClaw

AI Assistant

Reads and curates gmail, drafts replies, manages contacts.

chattpc26

OpenClaw

Conference Chair Assistant

Tracks, organizes form submissions (talks and session proposals), produces reports, sends notifications via Slack and email.

The stack

Docker on the DGX Spark: one container runs OpenClaw, a second runs Hermes, and a third runs vLLM serving `Qwen3-Coder-Next-FP8` at **~50 tps** — where most models on the Spark manage a dozen or two. Agents use remote models.

gandalf

Hermes

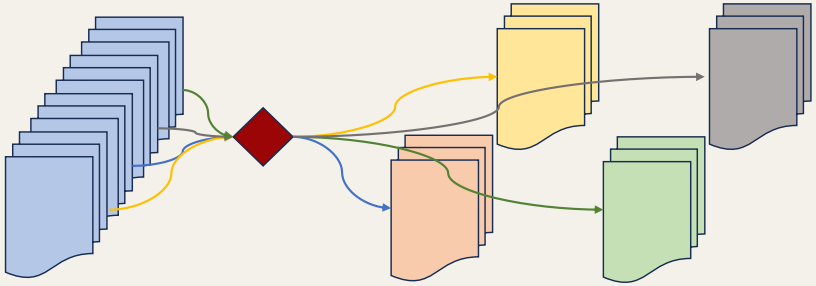
Inbox assist

Reads gmail, drafts replies, manages contacts.

PART 1 · THE SYSTEM

Chattpc26: an Agent to (Help) Organize a Major Conference

Curate 100-150 Lightning Talk Proposals by Session (Daily)



Review Lightning Talk Abstracts, Send Approvals (semi-weekly)

	Screen	Notified	Author	Institution	Title
Dupe		n/a	Emma Boucharc	Carnegie Mellon	AI and Experimental Automation
	✅	2026-05-19	Austin Cheng	University of Toronto	Nahual: A Sequence Model for Language and Atoms
	✅	2026-05-19	Pete Mendygral	HPE	DragonHPC: a High-performance Distributed Runtime for AI-coupled HPC Workflows
	✅	2026-05-19	William Merrill	Toyota Technological Institute / Ai2	Olmo Hybrid: From Theory to Practice and Back
Dupe	✅	2026-05-19	Samantika Sury	Hewlett Packard Enterprise	Macroheterogeneity: Enabling Hybrid HPC and AI workflows
		n/a	Samantika Sury	Hewlett Packard Enterprise	Macroheterogeneity: Enabling Hybrid HPC and AI workflows
	✅	2026-05-19	Padma Apparao	Intel Corporation	Scaling MoE Models to Exascale with Open Software on Aurora
	✅	2026-05-19	Edward (Ted) Se	Northwestern University	Human-in-the-Loop and Mixed Acceleration for Next-Generation Catalysis
	✅	2026-05-19	Padma Apparao	Intel Corporation	Scaling MoE Inference with vLLM on Aurora
	✅	2026-05-19	Massimiliano Lu	Oak Ridge National Laboratory	Scalable Heterogeneous Graph Learning for Grid AI Foundation Models
	✅	2026-05-18	Robert Underwo	Argonne National Laboratory	Best Practices for Scientific Workflows
Fixed	👍	2026-05-23	Zain Asgar	Gimlet Labs	Efficient and Scalable Agentic AI with Heterogeneous Systems
	✅	2026-05-18	Vineeth Gutta	NVIDIA	Prefill Here, Decode There: Disaggregated LLM Serving Across GPUs and LPUs
	✅	2026-05-30	Sean Shahkarar	Northwestern University	Bridging the computing continuum
Declined	✅	2026-05-30	Venkatram Vish	Argonne National Laboratory	ALCF Inference Service: Facility-Scale AI Services for Science
	❓	2026-05-30		City College of Chicago	Anti- deepfake and algorithm chain design

Semi-Weekly Slack Briefings for Program Committee

ChatCeC VIP APP 11:00 AM

We've got a modest uptick in activity with six new lightning talk proposals across four tracks—SHAR and SOFT each pulled double duty with two submissions apiece, while MAPE and EVAL added one each. No new tutorial interest beyond the single AI4S inquiry, and WG and BOF slot requests remain quiet at zero. The SHAR and SOFT surge suggests renewed energy in those areas, so folks might want to keep an eye out for cross-pollination opportunities. Otherwise, all quiet on the coordination front—great time to review pending proposals or reach out to submitters for clarifications if needed.

[ChatTPC26] TPC26 Submissions Summary

Lightning Talks: 6 new (108 total)
- SHAR: 2 new (30 total)
- MAPE: 1 new (11 total)
- SOFT: 2 new (22 total)
- EVAL: 1 new (10 total)

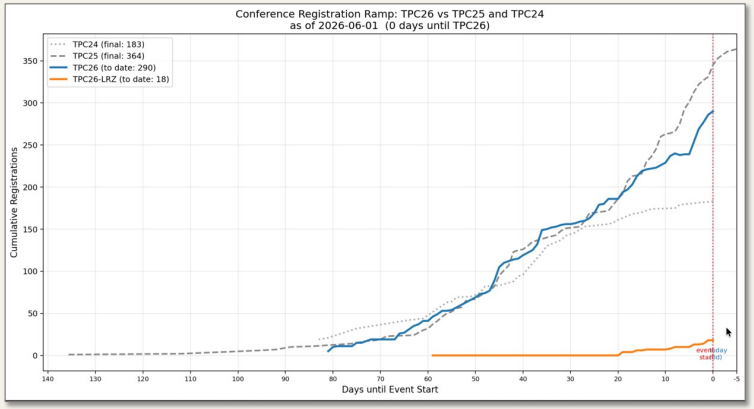
No new talks today for FAST (7 total), APPS (16 total), SAFE (3 total), or WORK (9 total)

AI4S Tutorial talks: 1 new (24 total)
WG Slots: 0 new (10 total)
BOF Slots: 0 new (13 total)

Lightning Talks	108	
APPS	16	
EVAL	10	
FAST	7	
MAPE	11	
SAFE	3	
SHAR	30	*Overflow risk*
SOFT	22	*Overflow risk*
WORK	9	

Breakouts	Groups	Slots
WG Sessions	10	17
BOF Sessions	13	13

Daily Registration Curve (Comparison w/ prior 2 years)



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PART 1 · THE SYSTEM

The Design Charter

Five principles, fixed before any external service was connected.

1	Identify the threat model	Address each identified threat surface
2	Alignment	Encode behavioral invariants structurally, not by trusting the model (plus item 5)
3	Separation of responsibilities	Code for procedure; the model for judgment
4	Containment	Network, execution, filesystem, config isolation in place first
5	Trust but verify	Consequential actions pass through a review layer (a separate agent) that learns

PART 1 · THE SYSTEM

The Threat Model

01 **Prompt injection**

LLMs do not separate instructions from data — a structural vulnerability

02 **Lateral movement**

A reachable sandbox becomes a pivot into your LAN or other nodes

03 **Runaway actions**

“Helpful” un-bounded: asked to diagnose, it tries to fix

04 **Credential exfiltration**

Secrets in the sandbox leak via unrestricted outbound HTTP

05 **Registry poisoning**

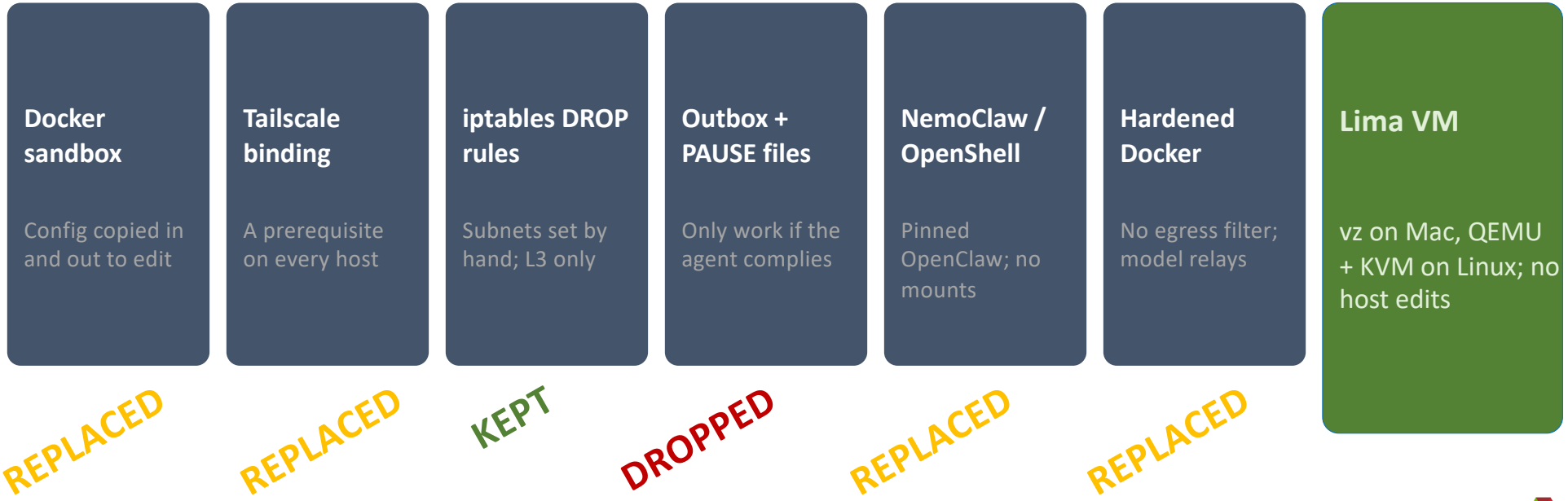
Third-party skills are code that runs in your sandbox

NOT HYPOTHETICAL **ClawHavoc** poisoned ~20% of the skill registry · **ClawJacked** (NIST CVE-2026-25253) exposed 135,000+ gateways bound to `0.0.0.0`

Seven Containment Strategies

59 Claude Code Sessions

Goal: explore how agents collaborate and interact, and understand the costs, risks and trade-offs of containment. Running two gateways, OpenClaw and Hermes, separates platform-specific issues from general ones.



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Six costs show up in every approach

Model access breaks at each boundary

Loopback-bound argo-shim needed 9 socat relays; OpenShell refused the Argo route; Hermes wants the OpenAI API shape.

Editing agent files gets harder

Direct in Docker, then sync scripts and sshfs under OpenShell, then gone in the VM: agents edit their own SOUL.md.

The substrate sets the pace

NemoClaw pinned OpenClaw at 2026.7.1, so native cron arrived too late to retire the old scheduler.

NemoClaw-specific

Containment hides failures

Mounts went stale while reporting healthy; 40 of 40 checks passed while a toolchain was missing.

Every integration is a hole

Slack, Google and Telegram each need egress policy, OAuth state and sometimes CA trust, reapplied after rebuilds.

Scaffolding for the scaffolding

Each local fix was reasonable; together they made a system hard to reproduce or hand to anyone else.

Four lessons for a contained agent lab

1 **A VM is the cleanest boundary**
A separate kernel, at the cost of host-side editing and a hypervisor dependency.

2 **Keep the model path outside containment**
It is the one hole every agent needs; rebuilding it per substrate was the largest source of complexity.

3 **Vendor containment must not pin the framework**
Otherwise the platform's own simplifications arrive too late to use.

4 **Verify from the inside, negatively**
A squid 403 and a refused proxy bypass prove more than any green status line.

https://github.com/cecat/bootstrap



Where bootstrap lands

- Lima VM: vz on macOS, QEMU + KVM on Linux
- No mounts, no inbound ports
- Host already reaches the model
- Agents edit their own files
- Telegram comms w/ humans
- Slack and Google as add-ons

CONCLUSION

Three Platform-Independent Thoughts

01

Caveat: This distinction is rapidly fading, with much more capable models internally using deterministic tools.

Determinism belongs in code

LLMs are non-deterministic. Give them skills and MCP access; skills using scripts where results must be exact. Don't ask a non-deterministic system to do a deterministic task — it will do it differently each time, often wrong.

02

Trust but Verify

Having models check one another makes good sense as part of a containment strategy, at minimum during the initial exercise of new duties.

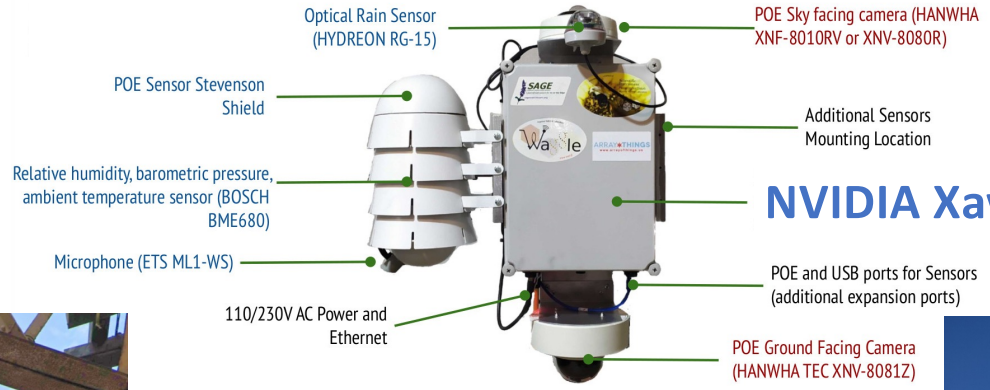
03

Memory and context = interesting and important science

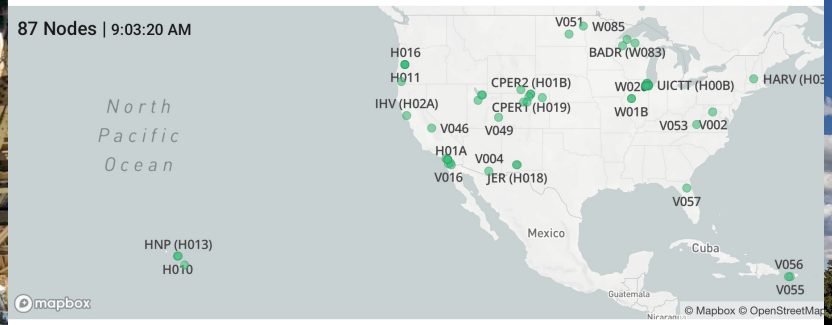
Balance token cost and performance against context cluttered with stale norms — and against context too thin, where the agent wakes up unaware of what it already did.



Embodied Agents



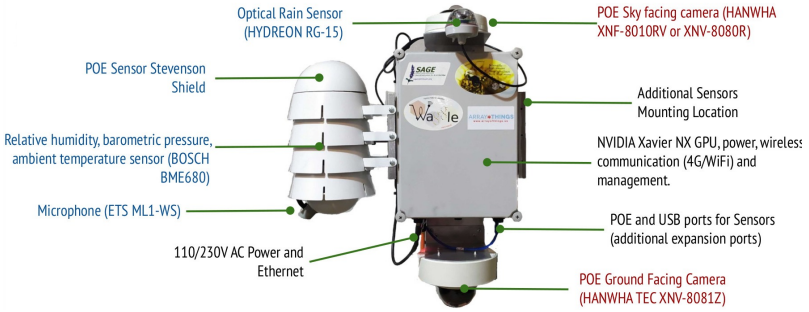




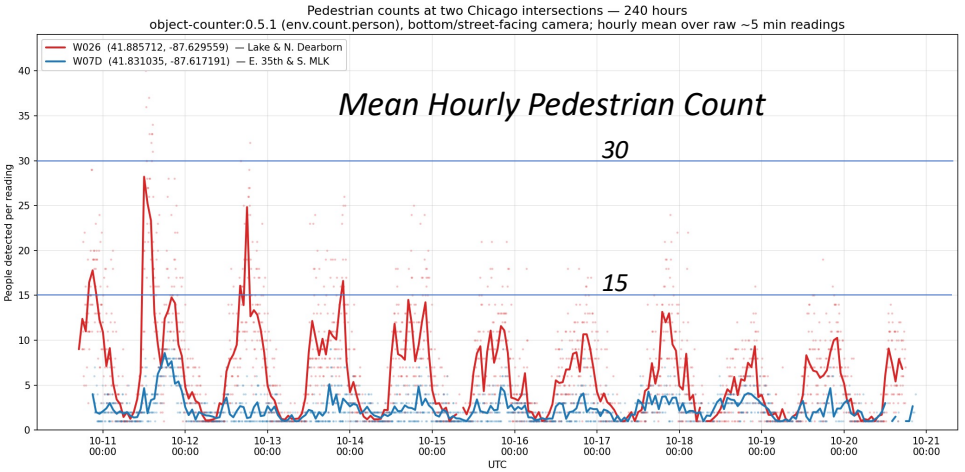


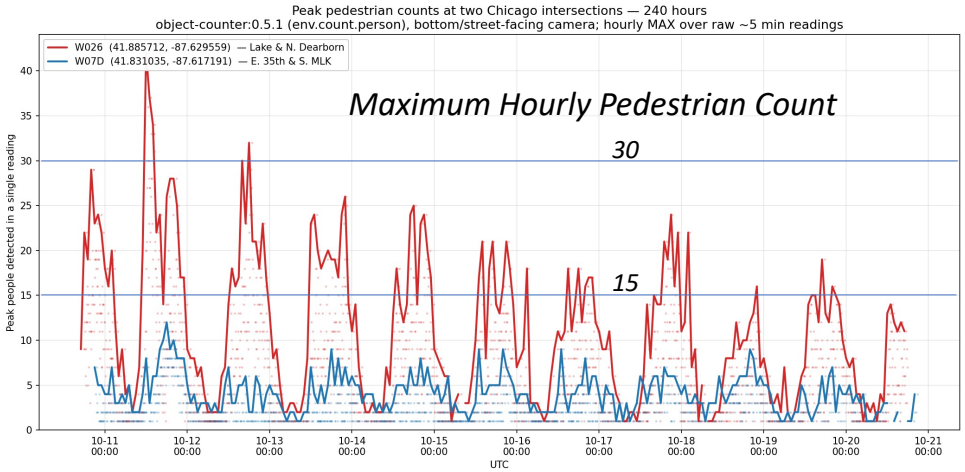
Embodied Agents





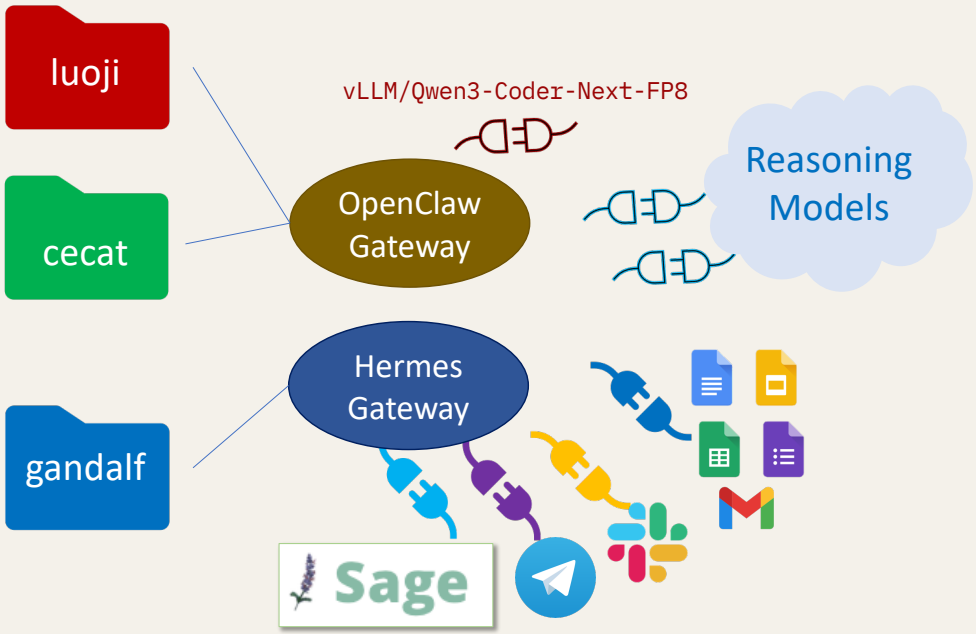






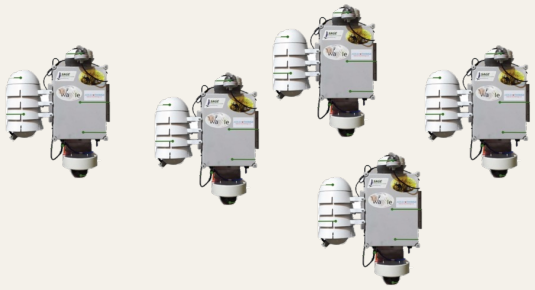
Pedestrian count – 240 hours at 5 minute intervals.

Embodied Agents



MCP access to:

- Central Data Repo
- Individual Nodes



Node Services:

- Read Live Data
- Run persistent “plug-ins”
  - Event/condition ”triggers” to actuate (local or remote, e.g, computation)

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Agents, MCP, and *Actuation*

CeC 11:43 AM
@ChatCeC show me several downward-facing camera images from Lake & Dearborn

ChatCeC VIP APP 11:44 AM
[LuoJi] Pulling the last several hours of bottom camera images from W026.

Six hourly images. Let me build the proxy links and verify they download.

All six downloaded successfully — all around 1MB each. Here are the bottom camera images from *W026 (Lake & N. Dearborn, Chicago)* spanning the last 6 hours:



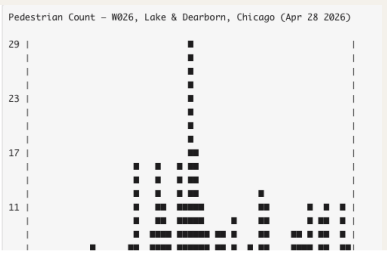
CeC 11:45 AM
@ChatCeC how many people do you see in each image? I'd like a time-series person-count for these images as a comma-delimited vector

ChatCeC VIP APP 11:46 AM
[LuoJi] I'd love to do that, but unfortunately I can't analyze the images — the image model isn't configured in this environment (it's set to a dummy API key). I can confirm the images downloaded successfully and are ~1MB each, but I can't count people in them.

Two options:

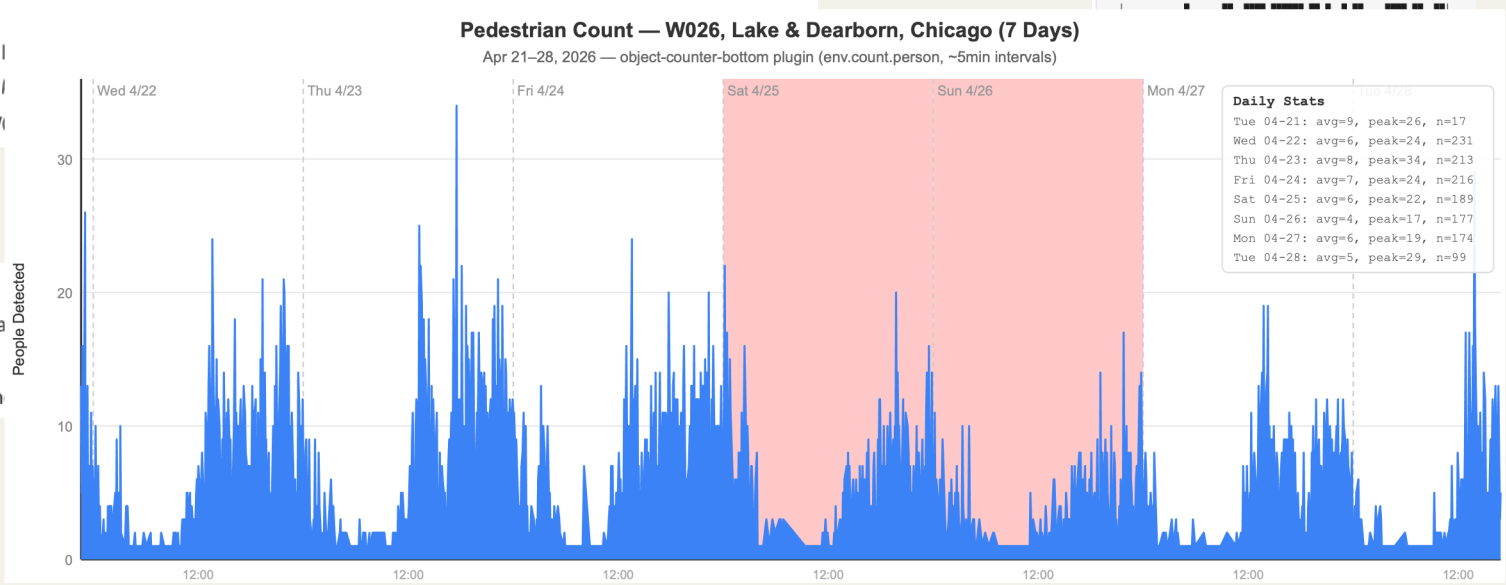
1. *You could eyeball the 6 images* from the links I
2. *There might be a SAGE plugin already running*

There's an object counter plugin running on W



CeC 11:57 AM
@ChatCeC Give me the chart for the past 7 da

ChatCeC VIP APP 11:58 AM
[LuoJi] Pulling 7 days of data and generating th



So **W026 and W07D are the only two with usable pedestrian time series**, and W026 is substantially the better one.

Worth noting this data is *not* blocked by the 403 problem — these are published numeric measurements rather than protected image files, so the inference output is fully readable even though the frames it came from aren't. That makes the object-counter a practical workaround for pedestrian analysis while your image access request is pending.

Want me to plot W026's pedestrian counts over the 240-hour window, or compare its diurnal pattern against W07D?

* Sautéed for 2m 17s

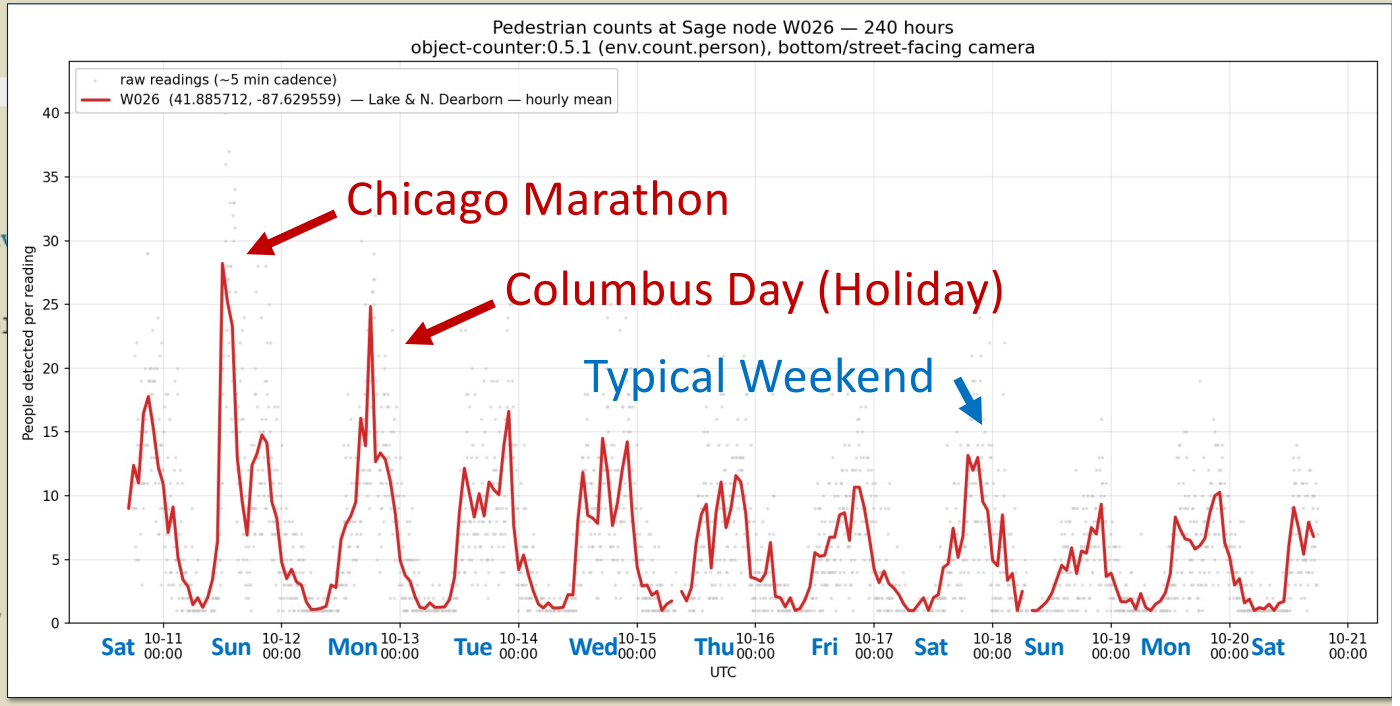
❯ yes, justr plot w026

● Write(plot_person_w026.py)
  └ Wrote 69 lines to plot_person_w026.py

```
     1 #!/usr/bin/env python3
     2 """Plot 240h of pedestrian counts (env
     3
     4 import csv
     5 from datetime import datetime, timedel
     6
     7 import matplotlib
     8 matplotlib.use("Agg")
     9 import matplotlib.dates as mdates
    10 import matplotlib.pyplot as plt
    … +59 lines
```

Read 1 file, ran 1 shell command

● The hourly-mean line is buried under the raw



"The hourly-mean line is buried under the raw noise. Let me fix visibility."

APPENDIX

References & Sources

RESEARCH

Zverev et al., “Can LLMs Separate Instructions From Data?” ICLR 2025.

Hong et al., “MetaGPT,” ICLR 2024 (oral). arXiv:2308.00352.

Qiu et al., “Blueprint First, Model Second,” arXiv:2508.02721, 2025.

Kapoor et al., “The 2025 AI Agent Index,” MIT, arXiv:2602.17753.

INCIDENTS & PLATFORMS

ClawHavoc skill-registry poisoning (Koi Security, Feb 2026).

ClawJacked — CVE-2026-25253 (SecurityScorecard STRIKE).

Hermes Agent — Nous Research (MIT). Cron, skills, memory docs.

NVIDIA NemoClaw + OpenShell — developer.nvidia.com (Apache-2.0).

Sage Project: www.sagecontinuum.org